UNIT 1 · STANDARD 6.NS.4

Least Common Multiple

Lesson 1-3

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

Multiples of 4: 4, 8, 12, 16, 20 — found by multiplying 4×1, 4×2, 4×3, ...

Multiple

What you get when you multiply a number by 1, 2, 3, and so on.

Multiples of 4: 4, 8, 12, 16, 20, 24. Multiples of 6: 6, 12, 18, 24. First match: 12 → LCM = 12

Least common multiple

The smallest number that two or more numbers both go into.

12, 24, and 36 are all common multiples of both 3 and 4

Common multiple

A number that two or more numbers both go into.

Skip count by 6: 6, 12, 18, 24, 30 — each jump adds 6

Skip counting

Counting by a number, like 2, 4, 6, to list its multiples.

$4 = 2 \times 2$, $6 = 2 \times 3$. LCM uses the highest power of each prime: $2^2 \times 3 = 12$

Prime factorization

Writing a number as prime numbers multiplied together. It helps you find the LCM.

Key Ideas & Notes

- Astronaut Reyes checks the oxygen system every 4 hours.
- Astronaut Chen checks the water recycler every 6 hours.
- They both start their shifts at 6:00 AM.
- When is the next time both astronauts will be checking their systems at the same time?
- Here are the first several multiples of 4 and 6 mixed together. Sort them — which are common multiples of BOTH 4 and 6, and which are multiples of only one?

Think About It

- How often does each astronaut check their system?
- What times will each astronaut be doing checks?
- When will both schedules line up at the same time?

My Notes

Turn & Talk — Discussion Points

Level 1 support Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Astronaut Reyes checks oxygen every 4 hours and Chen checks water every 6 hours, both starting at 6:00 AM. Why isn't the next shared check simply 10 hours later (4 + 6)?

Level 1 support

Start here: The next shared check is a common multiple of 4 and 6, not their sum.

Their checks line up at a ____ of both 4 and 6.

Sus revisiones coinciden en un ____ de 4 y de 6.

Adding 4 + 6 is wrong because ____.

Sumar 4 + 6 es incorrecto porque ____.

Word bank: multiple · common multiple · least common multiple · skip counting · schedule

Listen for: *Students reason that each astronaut repeats on multiples of their own number, so they only overlap at a common multiple, and adding 4 + 6 does not land on both schedules.*

Level 2

If Reyes checked every 4 hours and Chen every 8 hours, when would they next line up? Explain why the answer is not 4 + 8.

- They line up after ____ hours because ____.
- It is not 12 because the LCM of 4 and 8 is ____, since ____.

EXPLORE

You skip counted the multiples of 4 and 6. How did you find the least common multiple from your lists?

Level 1 support

Start here: The LCM is the first multiple that appears in both lists.

The multiples of 4 are ___ and the multiples of 6 are ___.
Los múltiplos de 4 son ___ y los múltiplos de 6 son ___.

The least common multiple is ___ because it is the ___ shared multiple.
El mínimo común múltiplo es ___ porque es el múltiplo compartido más ___.

Word bank: **multiple** **common multiple** **least common multiple** **skip counting** **smallest**

Listen for: *Students skip count 4, 8, 12 and 6, 12 and identify 12 as the LCM because it is the smallest number appearing in both lists.*

Level 2

Will the LCM of two numbers ever be smaller than the larger number? Justify using 4 and 6.

- The LCM is never smaller than the larger number because ___.
- For 4 and 6 the LCM is ___, which is ___ than 6.

CONNECT

Hot dogs come in packs of 8 and buns in packs of 6. How does the LCM help you buy the fewest of each so none are left over?

Level 1 support

Start here: The LCM of 8 and 6 is the smallest matching total of dogs and buns.

I need to buy ___ hot dogs and ___ buns because the LCM of 8 and 6 is ___.

Necesito comprar ___ hot dogs y ___ panes porque el mcm de 8 y 6 es ___.

That means ___ packs of hot dogs and ___ packs of buns.

Eso significa ___ paquetes de hot dogs y ___ paquetes de panes.

Word bank: **least common multiple** **multiple** **common multiple** **packs** **none left over**

Listen for: *Students compute LCM(8, 6) = 24, so 3 packs of hot dogs and 4 packs of buns give 24 each with nothing left over.*

Level 2

Why is the LCM the FEWEST you can buy and not just any common multiple like 48? Explain the trade-off.

- The LCM gives the fewest because ___.
- Buying 48 of each would also work, but ___.

REFLECT

How do you decide whether a real-world problem needs the LCM or the GCF?

Level 1 support

Start here: LCM answers 'when do repeating events meet'; GCF answers 'how to split into equal groups'.

I use the LCM when ____, but the GCF when ____.

Uso el mcm cuando ____, pero el MCD cuando ____.

The astronaut schedule needs the ___ because ____.

El horario de los astronautas necesita el ___ porque ____.

Word bank: **least common multiple** **common multiple** **multiple** **skip counting** **greatest common factor**

Listen for: *Students explain LCM is for repeating/lining-up events (schedules, packs) while GCF is for splitting into equal groups, and correctly tag the schedule as an LCM problem.*

Level 2

Write your own short word problem that needs the LCM, then explain why GCF would give the wrong answer.

- My problem is ____, and it needs the LCM because ____.
- Using the GCF would be wrong because ____.

Guided Examples

Example 1

What is the LCM of 3 and 5?

Solution: Multiples of 3: 3, 6, 9, 12, 15, ... Multiples of 5: 5, 10, 15, ... The smallest common multiple is 15.

Answer: A. 15

Example 2

What is the LCM of 6 and 8?

Solution: Multiples of 6: 6, 12, 18, 24, ... Multiples of 8: 8, 16, 24, ... The smallest common multiple is 24.

Answer: A. 24

Example 3

What is the LCM of 4 and 10?

Solution: Multiples of 4: 4, 8, 12, 16, 20, ... Multiples of 10: 10, 20, ... The smallest common multiple is 20.

Answer: A. 20

Write About the Math
The Writing Revolution

I can explain how I found the LCM using the words multiple, common multiple, skip counting, and least common multiple.

1. Kernel Sentence
subject + verb

Model: Multiple is what you get when you multiply a number by 1, 2, 3, and so on.
Múltiplo es lo que obtienes al multiplicar un número por 1, 2, 3, y así.

Write a kernel sentence about multiple. Use a subject and a verb.
Escribe una oración base sobre múltiplo. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion
because · but · so

Kernel: Multiple matters in math
Múltiplo importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*
Multiple matters in math because ____.
Múltiplo importa en matemáticas porque ____.

____

but *pero*
Multiple matters in math, but ____.
Múltiplo importa en matemáticas, pero ____.

____

so *entonces*
Multiple matters in math, so ____.
Múltiplo importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about multiple.
Di un hecho verdadero sobre multiple.

Multiple ____.

Question
Pregunta

Ask a question about multiple.
Haz una pregunta sobre multiple.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about multiple.
Muestra entusiasmo sobre multiple.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with multiple.
Dile a un compañero qué hacer con multiple.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

Their checks line up at a ____ of both 4 and 6.
Sus revisiones coinciden en un ____ de 4 y de 6.

Adding 4 + 6 is wrong because ____.
Sumar 4 + 6 es incorrecto porque ____.

The multiples of 4 are ____ and the multiples of 6 are ____.
Los múltiplos de 4 son ____ y los múltiplos de 6 son ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What is the LCM of 2 and 7?

A. 14

B. 7

C. 2

D. 9

Show your work:

2. Two buses leave the station at 8:00 AM. Bus A returns every 15 minutes and Bus B returns every 20 minutes. When is the next time both buses are at the station together?

A. 9:00 AM (60 minutes later)

B. 8:30 AM (30 minutes later)

C. 8:40 AM (40 minutes later)

D. 9:20 AM (80 minutes later)

Show your work:

Stretch Your Thinking
Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Hot dogs come in packs of 8 and buns come in packs of 6. What is the fewest of each you can buy so you have the same number of hot dogs and buns with none left over? Explain how the LCM solves this problem.

Sentence starter: The LCM of 8 and 6 is ____. I need to buy ____ packs of hot dogs and ____ packs of buns to get ____ of each. This works because ____.

Show your work:

Reflect — Exit Ticket

What is the LCM of 9 and 12?

A. 36

B. 108

C. 3

D. 72

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 14 *— Since 2 and 7 share no common factors (they are relatively prime), LCM = 2 × 7 = 14.*
2. **Try It 2:** A. 9:00 AM (60 minutes later) *— LCM(15, 20) = 60 minutes. Both buses return together 60 minutes after 8:00 AM, at 9:00 AM.*
3. **Exit Ticket:** A. 36 *— Multiples of 9: 9, 18, 27, 36, ... Multiples of 12: 12, 24, 36, ... The smallest common multiple is 36.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Multiple is what you get when you multiply a number by 1, 2, 3, and so on.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).